HAPPIEST HEALTH | DIGITAL TWIN CHALLENGE 2026

PranaTwin-T2D

A patient-specific digital twin for 2-hour glucose forecasting

A clinician-facing research prototype that synchronizes synthetic clinical history with five-minute telemetry and forecasts near-term glucose trajectories.

TWO-STREAM
FHIR + telemetry

15–120 MIN
Forecast horizon

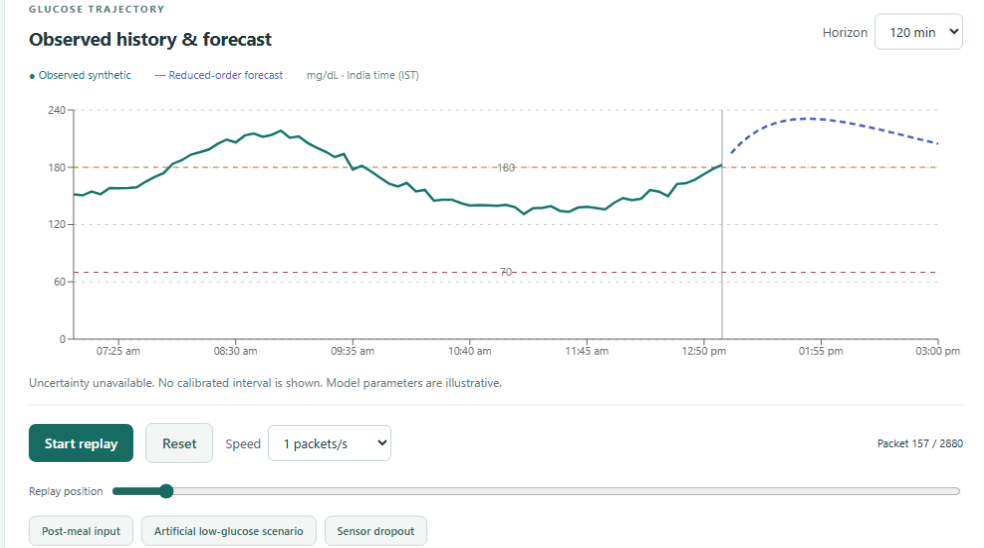
RUNNABLE PoC
API + web UI

Synthetic-data research prototype

RESEARCH USE ONLY

LIVE CONTROL ROOM

Synthetic patient PT-IND-001



183
Latest glucose

156
Twin baseline

120 min
Forecast horizon

Illustrative values from the synthetic replay environment

India's dysglycemia burden creates a clear case for earlier glucose visibility

ICMR-INDIAB estimates show a large population living with diabetes or prediabetes, alongside gaps in comprehensive target achievement. [1,2]

NATIONAL BURDEN

237M

adults with diabetes or prediabetes



101M diabetes

136M prediabetes

Adults • India • ICMR-INDIAB estimates

Population estimate

Adult estimates

CONTROL GAP

7.7%

achieved all three ABC treatment targets

Among people with self-reported diabetes in the ICMR-INDIAB-13 analysis. [2]

01 **Clinical history**

Diagnoses, laboratory values and medication context

02 **Daily physiology**

Glucose, activity, sleep and wearable signals

03 **Near-term forecast**

A 15–120 minute trajectory for clinician review

Use case: short-horizon glycemic excursion forecasting

Sources: [1] Anjana et al., ICMR-INDIAB-17, The Lancet Diabetes & Endocrinology (2023). [2] ICMR-INDIAB-13, The Lancet Diabetes & Endocrinology (2022).

PranaTwin links clinical history to an evolving glucose state and a two-hour forecast

The prototype synchronizes a synthetic FHIR record with time-stamped telemetry, then returns a forward glucose trajectory and threshold flags.

SYNTHETIC PROFILE PT-IND-001
58 years • T2D + hypertension / CKD context

BMI	WAIST	LATEST GLUCOSE	EHR BASELINE
27.1 kg/m²	98 cm	183 mg/dL	156 mg/dL

Historical EHR

FHIR R4 document
Prior glucose + diagnoses
Selected labs / medicine context

Dynamic telemetry

Five-minute replay packets
Glucose, HR, HRV, EDA, temperature
Activity and meal context

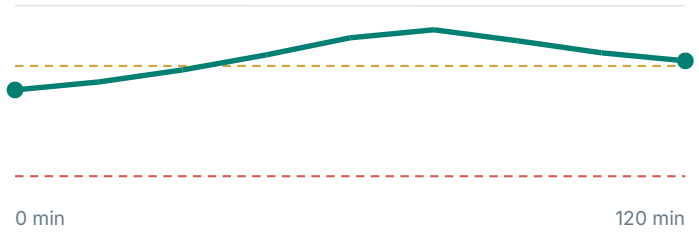
STATE SYNCHRONIZER

Virtual patient state

- G Observed glucose
- Q Meal / gut proxy
- B Adaptive baseline
- T Trend + signal quality

FORECAST SERVICE

15–120 minute glucose path

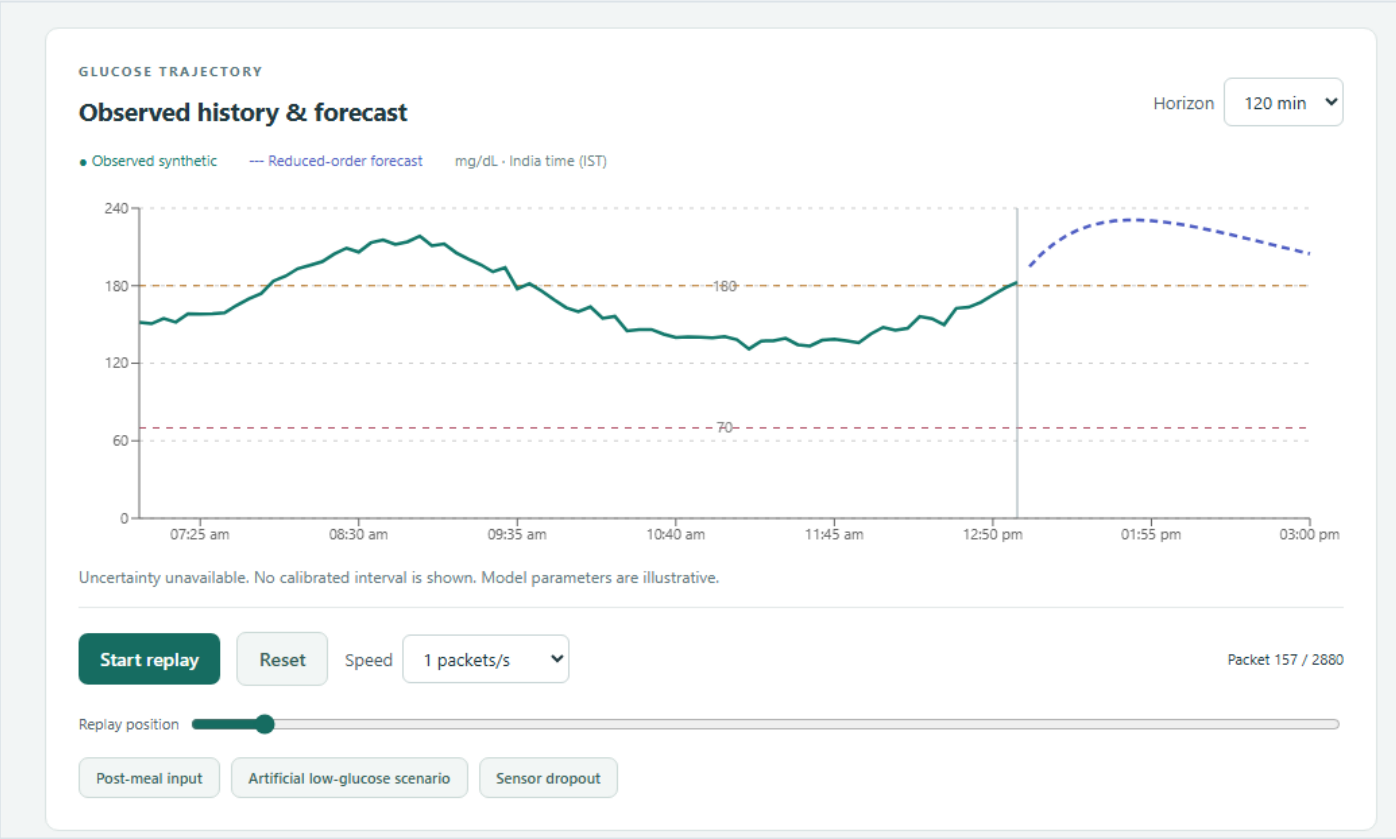


OBSERVED State updated on each packet → FORECAST 15–120 min trajectory + flags → CLINICIAN Review / compare / inspect

Glucose excursion thresholds: <70 mg/dL and >180 mg/dL [3]. Profile values shown above are fictional synthetic demo values.

The control room combines replay, forecasting and scenario analysis in one interface

A working React interface is served by the FastAPI backend; the screenshot shows synthetic patient PT-IND-001.



CLINICIAN WORKFLOW

Replay the timeline

Move through five-minute packets and inspect the current state.

Inspect EHR context

View the synthetic FHIR bundle, diagnoses and selected observations.

Review forecast flags

See the 15–120 minute trajectory and threshold crossings.

Compare scenarios

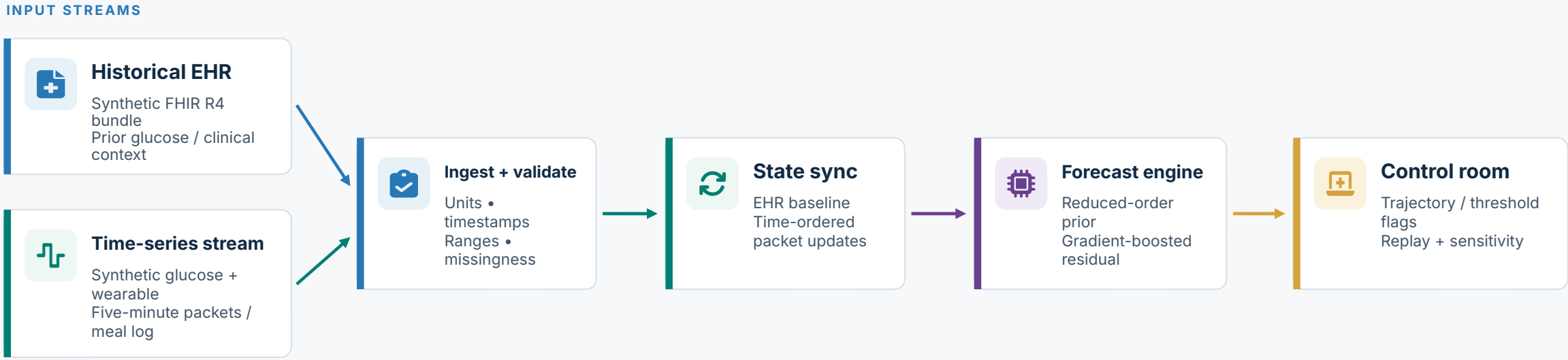
Change bounded meal/activity inputs from the same replay state.

REACT + TYPESCRIPT | FASTAPI + WEBSOCKET

Screenshot: synthetic profile PT-IND-001 from the local replay environment.

A six-stage pipeline transforms clinical context and telemetry into an inspectable forecast

The implementation separates data intake, temporal validation, state synchronization, prediction and clinician interaction.



IMPLEMENTATION STACK

React / TypeScript	FastAPI / Pydantic	Python / NumPy	scikit-learn	SQLite + Docker

Interoperability implementation follows the local FHIR document checks defined in docs/ingestion.md.

The forecasting engine combines a reduced-order glucose prior with learned residual correction

Model features include observed glucose dynamics, meal/activity proxies, selected wearable signals and EHR context. [5]

INPUT STATE

Glucose state
Observed glucose • trend • adaptive baseline

Meal / activity
Carbohydrate-gut proxy • activity level

Wearable context
HR • HRV • EDA • skin temperature

EHR context
BMI • waist • HbA1c • eGFR • medicine history

01 Reduced-order prior
Bergman/Hovorka-inspired compartment structure; bounded glucose / gut updates with illustrative parameters.

02 Learned residual
Gradient-boosted regression corrects the model prior using current-state and lagged features.

FORECAST OUTPUT

$$\hat{G}(t+h) = G_{prior}(t+h) + \hat{residual}(t+h)$$

15–120 min	Threshold flags	Research heads
Glucose trajectory	>180 / <70 mg/dL	Quantile + event outputs

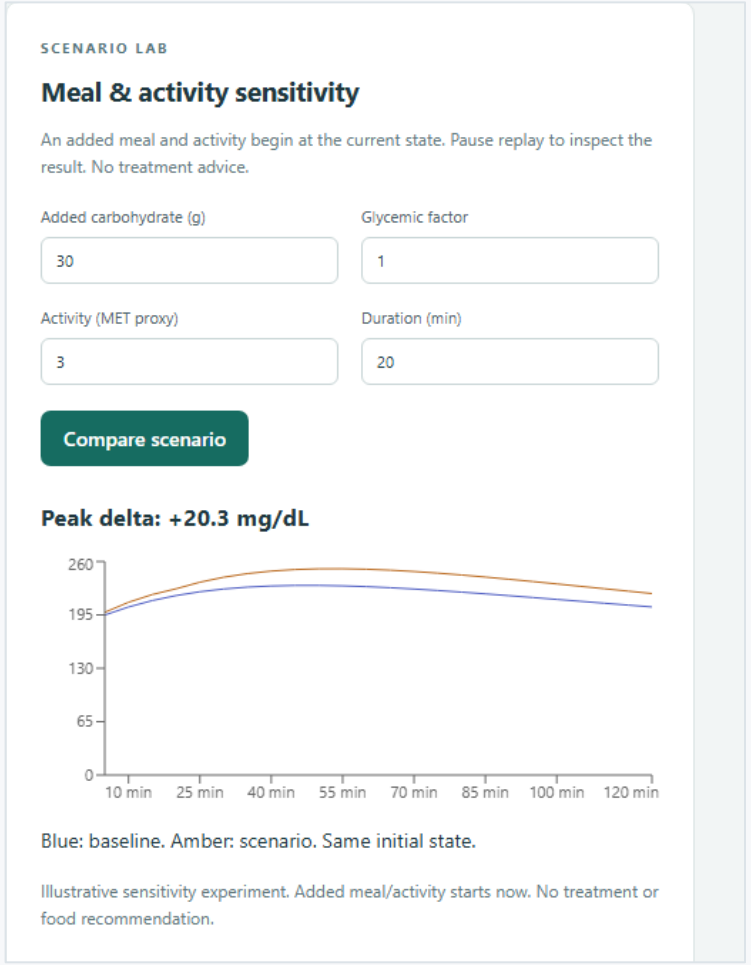
MODEL INTERPRETATION

Model parameters are fixed engineering priors; medicine history supplies bounded context factors to the forecasting layer.

Model implementation: engine/model.py, engine/hybrid.py, docs/model_methodology.md and reports/metrics.json [5].

Scenario lab compares projected trajectories from the same starting state

Meal and activity inputs are perturbed within defined bounds; the UI reports the difference between two simulated paths.



CONTROLLED COMPARISON

Same initial state

Current glucose, baseline, gut proxy and packet state are copied.

Baseline forecast

No added scenario input

Perturb input

Carbohydrate, glycemic factor, activity level and duration.

Scenario forecast

Same horizon; modified inputs

REPORTED COMPARISON

Peak glucose delta

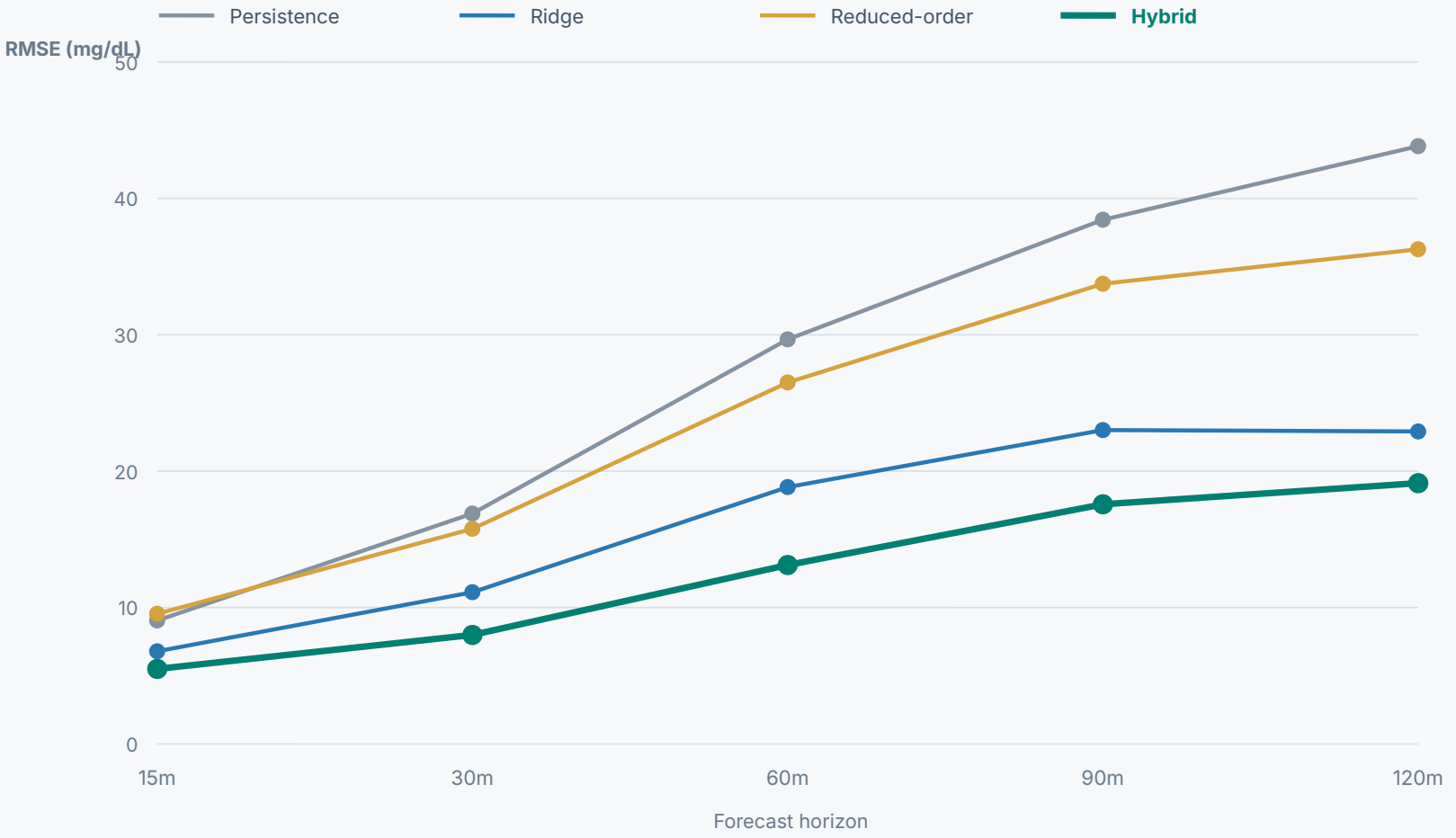
+20.3 mg/dL

Example output from the displayed synthetic replay

Scenario bounds and output fields are implemented in engine/model.py and engine/api.py. Sensitivity results are simulations, not causal estimates of real-world response.

Hybrid forecasting reduces RMSE against persistence across five synthetic test horizons

Within-profile chronological holdout; all methods are evaluated on the same eligible target windows at each forecast horizon. [5]



120-MINUTE HOLDOUT

19.12

mg/dL RMSE

vs 43.83 mg/dL

Persistence baseline

56% lower RMSE

Generated test tracks • 276 eligible targets

Source: reports/metrics.json, generated 20 Oct 2026. Evaluation data are synthetic; external-data validation is the next performance milestone.

Evaluation uses chronological splits and a forecast-horizon purge to limit temporal leakage

Three fictional profiles provide ten days of five-minute data each; the final time windows are held out for forecasting benchmarks. [5]

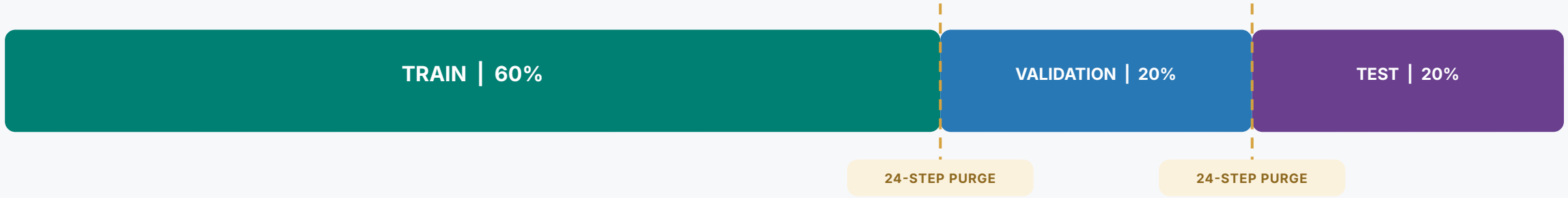
3
Synthetic profiles
Three distinct fictional contexts

8,640
Time-series rows
3 profiles × 2,880 packets

5 min
Packet cadence
Ten-day replay per profile

276
Test targets / horizon
Same eligible targets across methods

WITHIN-PROFILE TIME SPLIT



Forecast origin cadence
Test origins every 30 minutes; target horizon extends up to 120 minutes.

Training isolation
Preprocessing fitted on training data; validation partition reserved.

Reported metrics
RMSE, MAE and MARD; model comparisons use identical eligible targets.

Metrics use synthetic data; forecast windows overlap in time and are not an external validation cohort.

A lightweight local stack supports repeatable replay, inspection and audit

The repository packages the frontend, API, deterministic synthetic stream and offline evaluation artifacts in one reproducible project. [5]

SOFTWARE STACK

Clinician UI
React • TypeScript • Vite • Recharts

API + replay service
FastAPI • Pydantic • REST • WebSocket

Model + evaluation
Python • NumPy • scikit-learn • CSV / JSON

Local persistence
SQLite provenance • Docker Compose

DATA HANDLING & CHECKS

Synthetic by default
Three fictional profiles; no real names or ABHA identifiers.

Temporal / unit validation
Timezone-aware timestamps, mg/dL and monotonic packet order.

Signal quality indicators
Missing values, stale glucose and packet gaps are surfaced.

Demonstration consent gate
Local revoke / restore behavior protects routes and streams.

Next phase: ABDM FHIR profile conformance testing

Implementation inventory based on README.md, docs/ingestion.md, docs/security_privacy.md, docs/audit.md and source code [5].

A staged validation plan moves the prototype from synthetic replay to clinical evaluation

The next development milestones prioritize authorized data, externally held-out performance, and clinician workflow review.

STAGE 01 — DATA

External data readiness

- Obtain authorized CGM + wearable data
- Inspect subject IDs, license and units
- Build source-specific parser / provenance manifest

Deliverable: external-data benchmark dataset

STAGE 02 — MODEL

Model validation

- Run subject-held-out and forward tests
- Compare against persistence and simple models
- Assess error by horizon; calibrate event outputs

Deliverable: external validation report

STAGE 03 — CLINICAL

Clinical workflow study

- Agree endpoints with clinical collaborators
- Review alerts, usability and failure modes
- Obtain ethics / governance approvals before patient work

Deliverable: approved evaluation protocol

Current milestone: external-data performance evaluation

Future clinical evaluation is contingent on appropriate ethics, privacy and institutional approvals.

PRANATWIN-T2D | SUMMARY

One synchronized state. One forecast horizon. One clinician control room.

PranaTwin-T2D brings EHR context, five-minute synthetic telemetry, forecasting and controlled sensitivity testing into a single runnable research prototype.

56%
Lower 120-min RMSE
vs persistence on synthetic holdout

8,640
Synthetic packets
3 fictional profiles • 10 days each

15–120m
Forecast horizon
Updated from replayed state

WHAT THE PoC DEMONSTRATES

Two-stream synchronization
Synthetic FHIR history plus evolving telemetry

Working forecast workflow
Replay, prediction, thresholds and scenario comparison

Measured benchmark
Common held-out targets across four model baselines

NEXT MILESTONE

External-data validation

Subject-held-out performance • calibration • clinician review

Research prototype using synthetic data | Not for clinical use

Appendix | Complete RMSE and MAE benchmark by forecast horizon

Error is measured in mg/dL across 276 eligible held-out targets per method and horizon. Lower values indicate closer predictions.

METHOD	15 MIN	30 MIN	60 MIN	90 MIN	120 MIN
Persistence • RMSE	9.05	16.89	29.66	38.43	43.83
Ridge • RMSE	6.79	11.12	18.83	23.01	22.91
Reduced-order • RMSE	9.54	15.77	26.50	33.74	36.27
Hybrid • RMSE	**5.50**	**7.99**	**13.12**	**17.57**	**19.12**
Persistence • MAE	6.71	12.41	21.55	28.66	33.42
Ridge • MAE	5.07	8.24	13.55	17.38	17.54
Reduced-order • MAE	7.13	11.91	20.26	26.95	29.84
Hybrid • MAE	**4.11**	**5.86**	**9.70**	**12.98**	**14.25**

Evaluation protocol **Within-profile chronological 60/20/20 split with 24-step boundary purges; test origins every 30 minutes.**

Source: reports/metrics.json and reports/predictions.csv (generated 20 Oct 2026). Synthetic holdout results; test windows overlap in time.

Appendix | Selected references and implementation map

Selected clinical references and repository implementation evidence.

REFERENCES

[1] **Anjana et al.**
ICMR-INDIAB-17. Metabolic non-communicable disease health report of India. The Lancet Diabetes & Endocrinology, 2023.

[2] **Anjana et al.**
ICMR-INDIAB-13. Diabetes treatment target achievement in India. The Lancet Diabetes & Endocrinology, 2022.

[3] **Battelino et al.**
Clinical targets for continuous glucose monitoring data interpretation. Diabetes Care, 42(8), 2019. DOI: 10.2337/dci19-0028.

[4] **HL7 FHIR R4**
Documents / Bundle specification. https://hl7.org/fhir/R4/documents.html

[5] **PranaTwin-T2D repository**
README, model methodology, source register, engine code and reports/metrics.json. Build artifacts dated 20 Oct 2026.

IMPLEMENTED IN THE POC

DATA LAYER	**Synthetic FHIR R4 + five-minute replay**	engine/data.py • engine/ingest.py
STATE ENGINE	**Chronological synchronization + bounded state**	engine/model.py
PREDICTION	**Reduced-order prior + gradient-boosted learner**	engine/hybrid.py • engine/production.py
INTERFACE	**Replay, patient/FHIR view, scenario lab, evaluation**	frontend/src/App.tsx
EVIDENCE	**Held-out metrics, predictions, test artefacts**	reports/metrics.json • reports/predictions.csv

Model methodology: docs/model_methodology.md • Evaluation: reports/metrics.json

All external URLs and implementation details are listed in the repository source register [5].